

Overall Project Explanation

CyberShield is an interactive cybersecurity simulation platform.

It demonstrates how common attacks work and how cryptographic or network security defenses stop them.

The user first logs in or registers. Then they enter the command dashboard. They choose a scenario, click Launch, observe attack logs and alerts, then click Defend to activate protection.

The project includes four scenarios:

1. Bank Transaction MITM.
2. Password Brute Force.
3. Packet Sniffing.
4. Replay Attack.

The platform uses:

- React frontend.
- Flask backend.
- SQLite database.
- JWT authentication.
- AES, RSA, SHA-256, digital signatures.
- Flask-SocketIO for real-time updates.

Demo Flow For Exhibition

1. Open frontend.
2. Register or login.
3. Show dashboard.
4. Explain navbar logo:
 - a. Blue means neutral.
 - b. Red means attack running.
 - c. Green means threat neutralized.
5. Select Bank MITM.
6. Click Launch.
7. Show red popup and hacker terminal.
8. Click Defend.
9. Show green popup and defender console.
10. Repeat quickly with brute force, packet sniffing, and replay attack.
11. Explain backend concepts and crypto concepts.

